Challenge: Precision Parking

Points: 150 • Estimated Time: 45 Minutes

Challenge Scope

The robot must navigate a marked course and park completely inside a designated 30cm x 30cm square zone.

Instructions & Engineering Hints

• Welcome to the **Precision Parking** challenge! Your mission is to program your robot to navigate a designated course and come to a complete stop entirely inside a 30cm x 30cm target parking zone.

• **Strategic Approach:** Break the run into distinct phases: leaving the start area, following or measuring the path, and executing the final parking maneuver. Relying solely on blind time-delays can be unreliable due to battery drain and wheel slip.

• **Sensor Placement Tips:** Consider equipping your robot with distance sensors (like ultrasonic or infrared) or line-following sensors if the course utilizes guidelines. Think about *where* on the chassis these sensors should be placed—front-mounted sensors help detect obstacles or walls ahead, while downward-facing sensors can help count grid lines or detect the boundary of the parking zone.

• **Structural Considerations:** Ensure your robot has a stable center of gravity so it doesn't tip over during sharp turns. Make sure your wheels have good traction to ensure consistent stopping distances when reversing or braking.